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Admission to the master's thesis

Dear Mr. Panza,

the Examination Board for the master's program Cognitive Neuroscience – has positively decided about your request. You will be licensed according to the § 9 of the examination regulations of the master's program for education sciences for the master's thesis.

According to your suggestion you will be assigned to the following examiners:

1. examiner (carer): **Univ.-Prof. Dr. Radoslav Cichy**

2. examiner: **Dr. Marleen Haupt**

As a suggestion from your 1. examiner, Univ.-Prof. Dr. Cichy, the jury for the master's program SCAN announces the following theme for your master thesis:

Effect of Masking on Putative Recurrence-Dependent Images

The master thesis processing time is 22 weeks.

The deadline is on **22 February 2024**.



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The attached statement or similar wording is yours on all copies
Include the master's thesis as the last page.

Sincere regards,
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Declaration

Herewith I affirm that I wrote the work in hand autonomously and never have used another source or resource as declared.

Berlin 19.02.2024

Danilo Peres

.....
Date and signature

Effect of Masking on Putative Recurrence-Dependent Images

Thesis submitted in partial fulfilment of the requirements for the degree
Master of Science

Master's degree program Cognitive Neuroscience Berlin (MCNB)
Department of Education and Psychology
Freie Universität Berlin

First Supervisor: Prof. Dr. Radoslaw Martin Cichy
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Submitted by
Davide Panza

Berlin

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Abstract

Traditionally, scientists viewed object recognition as a cognitive process carried out via rapid feedforward processing along the ventral visual pathway. However, this pathway also involves recurrent processing through lateral or feedback connections between areas, which emerges later in time compared to feedforward processing. Growing evidence suggest that this recurrent processing may play a crucial role in recognizing images under challenging conditions, such as occlusion or partial visibility. Recent studies have utilized a feedforward Convolutional Neural Network (CNN) to model the feedforward connections along the ventral visual pathway, introducing a novel computational approach for identify images with high and low reliance on recurrent processing. Their findings revealed that images recognized correctly by humans, but not by the CNN, necessitated additional time for accurate decoding from whole-brain neural activity, implying an involvement of slower recurrent processing in the recognition of these images. Expanding on these findings, we investigated how disrupting recurrent processing through backward masking affects the recognition and neural decoding of images previously classified as recurrent-dependent (Challenge) or feedforward-only (Control), based on the gap in object recognition performance between humans and a feedforward CNN. Moreover, we examined whether CNNs architectures that implement recurrence-like features, such as deeper models or with recurrent connections, demonstrate an enhanced ability to capture human behavior. Our findings indicate that the recognition and decoding accuracy of Challenge images are more affected by disruption of recurrent processing compared to Control images, suggesting a stronger reliance on such processing. Furthermore, we found that models incorporating recurrence-like features better capture human behavior, particularly for Challenge images. This may suggest that these models partially succeed in capturing human recurrent processing.

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Abstract

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1. Introduction

A remarkable ability of humans and other primates is their capacity to recognize objects with great ease, despite the considerable variability in their appearance (Cox et al., 2005; DiCarlo et al., 2012; DiCarlo & Cox, 2007). Scientists agree in identifying the ventral visual pathway, a network of hierarchically organized brain areas, as the locus where the neural processing leading to object recognition take place (Kravitz et al., 2013). This network includes extensive feedforward and recurrent connections (Felleman & Van Essen, 1991), each characterized by distinct temporal dynamics in their neural processing. In particular, previous studies have indicated that recurrent processing operates at a slower timescale compared to feedforward processing, typically exhibiting latencies ~ 50 ms longer on average (Lamme & Roelfsema, 2000; Wyatte et al., 2014).

Growing evidence supports the idea that these two pathways play distinct functional roles in object recognition, highlighting their diverse contributions for the recognition of easily identifiable images versus more challenging ones, like occluded or degraded images (Wyatte et al., 2014). Researchers found that the object identity of easy images could be decoded from human inferior cortex (IT) activity within ~ 150 ms after stimulus presentation (Rousselet et al., 2002; Thorpe et al., 1996). In contrast, the object identity of partial images (Tang et al., 2014) or occluded images (Rajaei et al., 2019) emerged ~ 50 ms later from IT activity, exhibiting a delay that fits well with the engagement of recurrent processing. Based on the observations of these differing temporal dynamics, researchers propose that feedforward processing alone suffices for achieving object recognition in easy images, whereas more challenging images require the additional involvement of recurrent processing (Kietzmann et al., 2019; Tang et al., 2018). Despite these findings, the distinct contribution of these two types of processing to object recognition remains controversial (Gilbert & Li, 2013; Kafaligonul et al., 2015).

To further investigate the role of recurrent processing in object recognition, Kar and colleagues (2019) recently employed a novel computational approach. They utilized AlexNet (Krizhevsky et al., 2012), a benchmark shallow feedforward Convolutional Neural Network (CNN), to model the feedforward segment of the ventral visual pathway. By leveraging the performance gap in image recognition between monkeys and AlexNet, they divided a set of images into two groups: easy-to-recognize (Control) and hard-to-recognize (Challenge). Challenge images were those correctly classified by monkeys but not by AlexNet, while

Control images were accurately classified by both. Their rationale was grounded in the assumption that AlexNet's misclassifications were attributed to its absence of recurrent connections, which they hypothesized to be crucial for the recognition of more challenging images. Remarkably, during neural decoding, they observed that accurate decoding of object information requires more time for Challenge images compared to Control images, displaying temporal dynamics consistent with the engagement of recurrent processing. To substantiate those findings, Kar et al. (2019) conducted an additional experiment, employing backward masking while presenting images to monkeys. Notably, backward masking has been shown to disrupt recurrent processing while leaving feedforward processing relatively unaffected (Fahrenfort et al., 2007). In line with their expectations, they observed a stronger impairment in the recognition of Challenge images following backward masking, confirming that Challenge images rely more on recurrent processing.

The generalizability of these results to humans has been investigated in studies that applied Kar's approach, using both synthetic images (Oyarzo et al., 2022) and more ecologically valid natural images (Boyanova, 2023). Aligning with Kar's findings, both studies reported a delayed emergence of object identity for Challenge images compared to Control images. However, these studies did not perform a systematic disruption of recurrent processing, a step that would provide converging evidence of the stronger reliance of Challenge images on this processing, corroborating the neural results.

In our study, we aimed to address this gap. Using a set of natural images previously divided into a Control and a Challenge group following Kar's approach (Boyanova, 2023), we first investigated how the disruption of recurrent processing, achieved through backward masking, affected the recognition of each group of images. Subsequently, we examined the similarities between the internal representations of various CNN architectures and the behavioral results using the Representation Similarity Analysis framework (Kriegeskorte, 2008). Through this step we aimed to explore how different CNN architectures capture human object recognition. Finally, we performed an EEG study to assess how disruption of recurrent processing affects the neural decoding of the two image groups.

In both the behavioral and EEG experiments, we gathered converging evidence indicating that disrupting recurrent processing has a stronger impact on Challenge images. This supports the hypothesis of a stronger involvement of this processing for the recognition of Challenge images, further indicating that Kar's results can be generalized to humans and natural images. Moreover, we found that CNN architectures implementing recurrent-like features, such as deeper models or with recurrent connections, exhibited better alignment

with human behavior, especially for Challenge images. This may suggest that adding recurrent-like features to CNNs allows them to partially capture human recurrence.

2. Material and Methods

2.1 Behavioral Experiment

2.1.1 Participants

30 healthy participants (21 female, mean age = 24.18, SD = ± 4.04) took part in the study. All participants had normal or corrected-to-normal vision and they had not previously participated in studies involving the same experimental stimuli. All procedures were approved by the ethical committee of Freie Universität Berlin and conducted in accordance with the Declaration of Helsinki. All participants provided written informed consent and received either €18 or research credit as compensation for their participation.

2.1.2 Experimental Stimuli

We utilized 212 images (**Fig. 1A**) sourced from ECOSET (Mehrer et al., 2021), a large-scale image dataset designed for human visual neuroscience, comprising 1.5 million images from 565 categories. Each of our images depicted one of five objects (bird, book, bottle, clock, elephant), with a varying number of images per object (bird = 44, book = 58, bottle = 46, clock = 38, elephant = 26). We centrally cropped the images to a square size of 224 x 224 pixels to ensure that all the images had the same dimension. The images were evenly divided into two groups (Control and Challenge), for a total of 106 images per group. The creation of these two groups was conducted in a previous study (Boyanova, 2023) following an approach introduced by Kar and colleagues (2019). Following this method, the gap in recognition performance between humans and AlexNet (measured in d' units) was used to identify images with a low (Control) or high (Challenge) reliance on recurrent processing. The Control group comprises images where the gap did not exceed 0.4 d' units, while the Challenge group includes images where human performance was at least 1.1 d' units greater than AlexNet performance.

Furthermore, in creating the two groups of images, the researchers ensured that each Control image had a Challenge counterpart displaying the same level of human performance. Only these matched pairs were included in the groups. This pairing procedure was performed

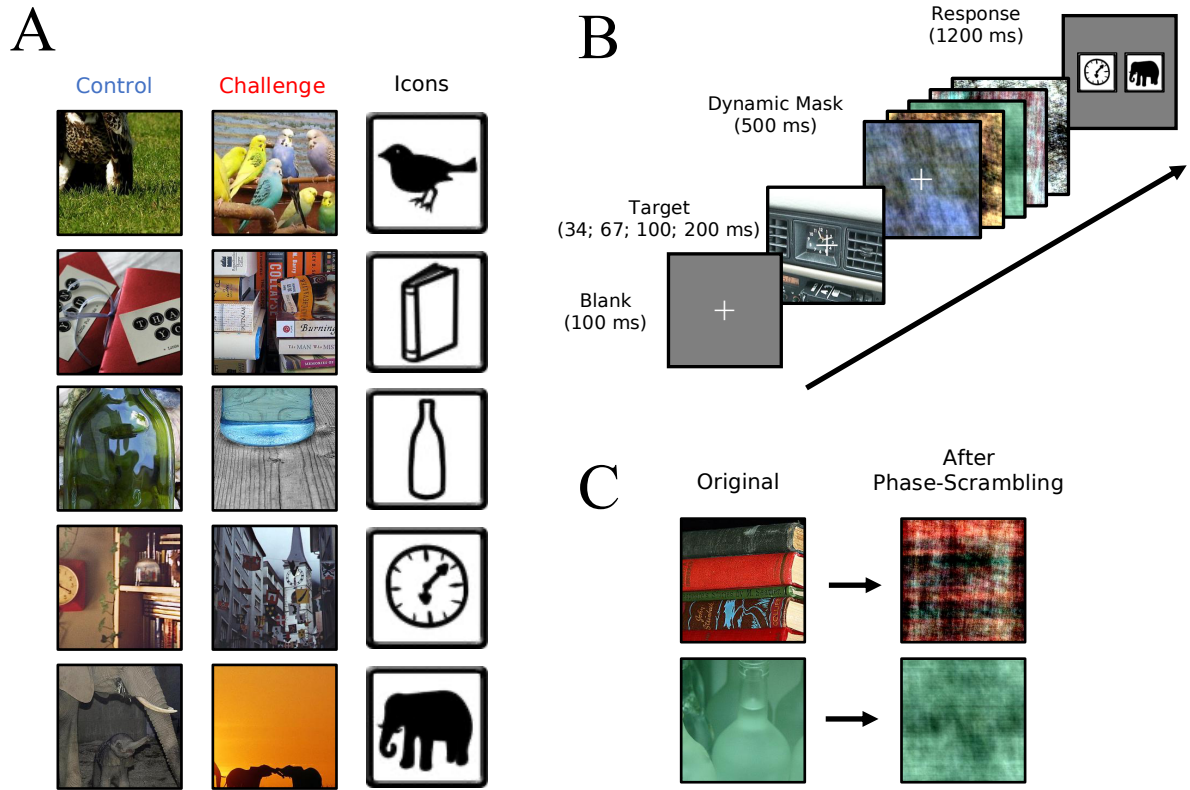


Figure 1. Behavioral Experiment Overview and Stimuli. **A)** A sample of the Control and Challenge stimuli used. The five COCO icons used in the binary discrimination task are depicted on the right, representing the objects featured in the images. **B)** A schematic overview of the behavioral experiment. **C)** An example of two images before and after phase-scramble transformation.

using the Kuhn-Munkres algorithm, a method that optimally matches elements from two sets, minimizing the overall cost of pairing (Frank, 2005). This step was necessary to prevent potential confounding effects arising from differences in human performance between the two image groups. Indeed, Kar's approach focuses on variations in model performance, not human performance, to identify images with different reliance on recurrent processing.

2.1.3 Experimental Design and Procedure

We implemented an experiment consisting of a binary object discrimination task (**Fig. 1B**). In this task, participants were required to indicate the identity of an object depicted in a target image by choosing between two alternatives.

In each trial, a grey screen with a fixation cross was initially displayed for 100 ms, then a target image appeared for a variable PT, followed by a dynamic backward mask for

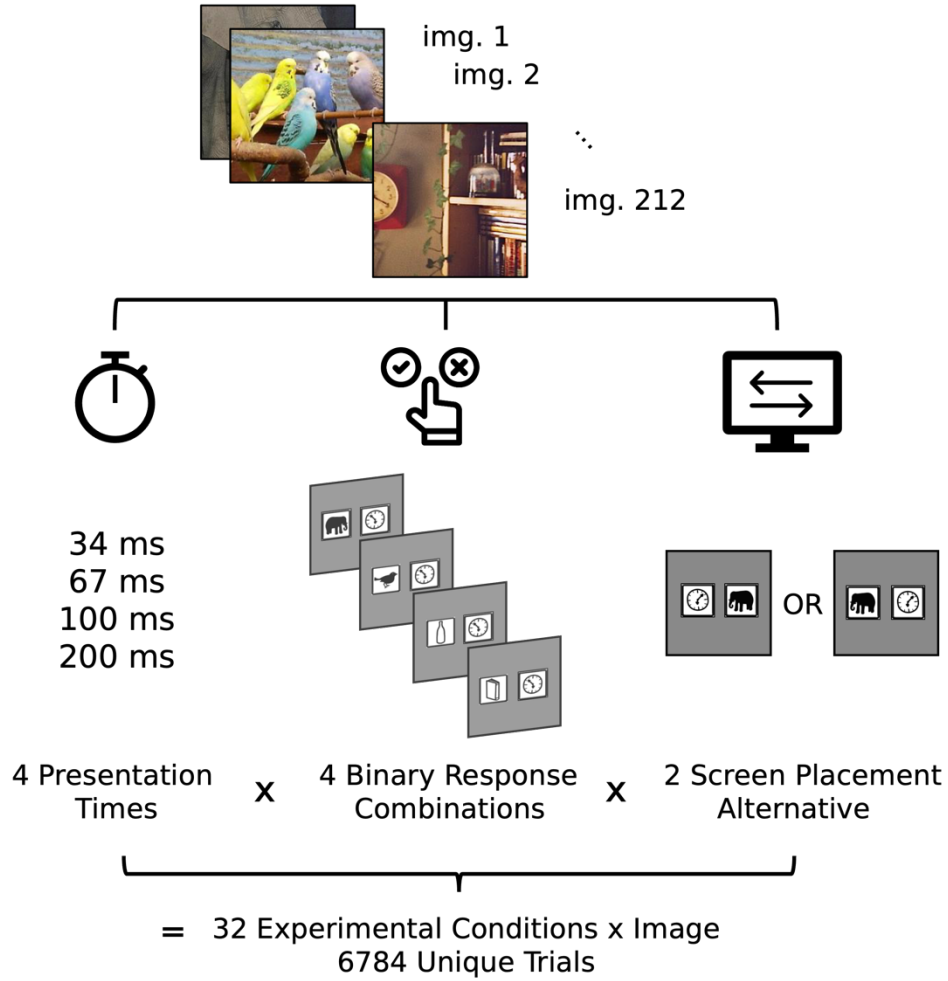


Figure 2. Overview of Trials Creations: This schematic illustrates the process of trial generation for the behavioral experiment. Each of the 212 images resulted in 32 distinct trials, formed by combining 4 implemented PTs, 4 binary response alternatives, and 2 screen placement options for the icons (right/left or left/right). In total, all the distinct trials created summed to 6874.

500 ms and a response window lasting 1200 ms. For each trial, a unique dynamic mask was created by randomly selecting five images from each object category, applying a phase-scramble transformation to them (**Fig. 1C**), and randomly rearranging the order of the images. The target images were presented at one of four different PTs (34, 67, 100, and 200 ms). Each phase randomized image of the backward masking was displayed in succession for 100 ms. During the response window, two COCO icons (Lin et al., 2015) representing the target and a distractor object (**Fig. 1A right**) were displayed on the screen. Participants were required to indicate which of these two objects was shown in the target image. For every image, four distinct response combinations were generated, each featuring the target object icon in combination with one of the four distractor object icon (**Fig. 2 middle**). To ensure an

equal probability of appearance on either side, we randomized the placement of the icons on the screen (**Fig. 2 right**).

In total, 6784 unique trials were created (**Fig. 2**), resulting from combinations of 212 target images, 4 target presentation times, 4 binary object discrimination tasks per image, and 2 icon positions (left/right or right/left). To manage the experiment within a reasonable timeframe, the total trials were pseudorandomized and divided into four sets. Regardless of the experimental condition (e.g. at different PTs), we ensured that each image was included the same number of times in each set. Every participant got assigned one set and completed a total of 1696 trials, which took about 1 hour and 15 minutes to finish. Prior to data collection, participants completed 20 practice trials to familiarize themselves with the task. A total of 50,880 trials were collected and submitted to further analysis. We used PsychoPy (Peirce et al., 2019) to design and conduct the experiment, and MATLAB (2021a) to perform the phase-scrambled transformation adapting a script developed by Hebart (2009).

2.1.4 Behavioral Analysis

We used d' as an indicator of participants' performance in recognizing individual images at different PTs. We computed a variant of d' using an one-versus-other image-level performance metric (Rajalingham et al., 2018), which estimates the discriminability of each image from each distractor object.

For each image, we initially calculated four hit rates, one for each of the distractors included in the binary task. This calculation involved determining the probability of correct responses, for trials including the same distractor-object. The four resulting hit rates were then averaged to obtain a single hit rate. The formula used was the following:

$$(2.1) \quad \text{Hit rate}_{\text{image}}^i = \frac{\sum_{j=1}^5 P_{\text{image}}^{i,j \neq i}}{4}$$

where P_c represents the probability of getting a correct response for an image of object i and all four distractors $j \neq i$. As ceiling or floor hit rate performances result in indefinite d' , we addressed this potential issue by implementing a $1/(2N)$ correction (Macmillan & Kaplan, 1985). Accordingly, hit rates equal to 1 were substituted using the following formula:

$$(2.2) \quad 1 - 1/(2n)$$

and hit rates equal to 0 were substituted using this formula:

$$(2.3) \quad 1/(2n)$$

where n represents the number of trials for the current image.

False alarm rates were calculated using the following formula:

$$(2.4) \quad \text{False alarm}^i = 1 - \text{avg}(\text{Hit Rate}_{image}^{j \neq i})$$

Finally, the d' was computed as follows:

$$(2.5) \quad d'_{image} = z(\text{Hit rate}^i_{image}) - z(\text{False alarm}^i_{image})$$

where z indicates normalization through z-transformation.

Moreover, we recorded participants' response times (RTs), which measured the time elapsed between the start of the response window and the press of the response key. Trials without responses (2.4% of all trials) were excluded from this analysis. Subsequently, RTs were normalized using z-transformation.

2.2 Computational Modeling

2.2.1 Models Architecture

We investigated six different CNN models (**Table 1**): AlexNet (Krizhevsky et al., 2012), VGG16 (Simonyan & Zisserman, 2015), ResNet50 (He et al., 2016), CorNet-z, CorNet-rt, and CorNet-s (Kubilius et al., 2018). We chose these models from the extensive CNN taxonomy to encompass widely used CNNs with diverse architectures. This includes shallow feedforward CNNs (AlexNet, CorNet-z), deeper feedforward CNNs (VGG16, ResNet50), or architectures incorporating recurrent connections (CorNet-r, CorNet-s). Aside from the implementation of different types of recurrent connections, we considered the addition of layers (deeper models) as a recurrence-like feature, as previous studies have shown that an unrolled recurrent network is equivalent to a deeper feedforward network (Liao & Poggio, 2016; Nayebe et al., 2018).

The choice of these CNN architectures enabled us to examine whether architectural differences contributed to varying the similarities between CNNs' internal representations and behavioral results. All these models were pre-trained on ImageNet and implemented in PyTorch (Paszke et al., 2019).

Table 1. Overview of Architectural Characteristics in Investigated CNN Models

<i>Models</i>	<i>Number of Parameters</i>	<i>Convolutional Layers</i>	<i>Pooling Layers</i>	<i>Fully-connected Layers</i>	<i>Connectivity Types</i>
AlexNet	61 100 840	5	3	3	feedforward
VGG-16	138 357 544	13	5	3	feedforward
ResNet-50	25 557 032	49	1	1	feedforward; skip
CorNet-z	2 071 656	4	-	1	feedforward
CorNet-rt	5 208 936	8	1	1	recurrent within block
CorNet-s	53 416 616	14	2	1	recurrent within block; skip

2.2.2 Representation Similarity Analysis

We employed Representation Similarity Analysis (RSA, Kriegeskorte, 2008) to assess the similarity between the internal representations of various CNN architectures and human performance. Within the RSA framework, various measurement modalities (e.g., EEG or fMRI recordings, behavioral data, etc.) are represented by a Representation Dissimilarity Matrix (RDM). This matrix stores pairwise distances between experimental conditions, serving as a common representational space that enables comparisons across different types of measurements.

To compare the internal representations of models with human behavior, we computed a similarity measure by correlating together behavioral and model RDMs, using Spearman’s rank correlation. We exclusively correlated the lower triangle of the RDMs (excluding the diagonal), as those are symmetrical.

2.2.3 Behavioral RDMs

From the results of the behavioral experiment, we computed the probability of selecting the target or the distractor for each target-distractor pair. This analysis was conducted independently for each experimental condition, resulting from the combinations of the two image groups (Control and Challenge) and four PTs. As a result, eight matrices were generated, each with dimensions of 106 x 5 (images x objects), containing the probability of each object being selected as a response in the respective images. From these matrices, we

constructed behavioral RDMs of size 106 x 106 (images x images), where each entry represented 1 minus the Pearson correlation between two 5-dimensional vectors.

2.2.4 CNN's RDMs

We computed separate RDMs for each image group at each layer of the implemented CNN architectures. We first inputted all images into the pre-trained CNNs and extracted the activation patterns from every layer of the models. The feature extraction was performed using the THINGSvision library (Muttenthaler & Hebart, 2021). We then conducted Principal Component Analysis (PCA) on the activation patterns, extracting the first 1000 components. This step aimed to standardize and equalize the activation sizes across layers.

To classify the object identities, we employed one-vs-all Support Vector Machine (SVM) classifiers, using the scikit-learn library (Pedregosa et al., 2011). Our training and testing procedures adopted a leave-one-out cross-validation approach. We used SVM classifications to build our RDMs, as SVM classifiers offer the capability to selectively weight only the relevant features. Similarly to the behavioral data, SVM outputs were stored in 106 x 5 matrices (images x objects), containing the probability scores of each object in the respective images. From these matrices, we employed the same methodology as in the behavioral results to derive RDMs.

2.3 EEG Experiment

2.3.1 Participants

5 participants (3 female, mean age = 28.5, SD = ± 6.13) took part in the experiment. All participants had normal or corrected-to-normal vision and they had not previously participated in any study using the same experimental stimuli. All procedures were approved by the ethical committee of Freie Universität Berlin and conducted in accordance with the Declaration of Helsinki. All participants provided written informed consent before conducting the study and received as compensation for their participation €12 euro per hour.

2.3.2 Experimental Stimuli

We selected a subset of images from the stimuli used in the behavioral experiment. This was done to reduce the recording time and avoid excessively long sessions. Our selection criteria aimed to maximize the observed effects from the behavioral experiment. As we observed a reduced d' for Challenge images at the 67 ms PT, we consequently retained

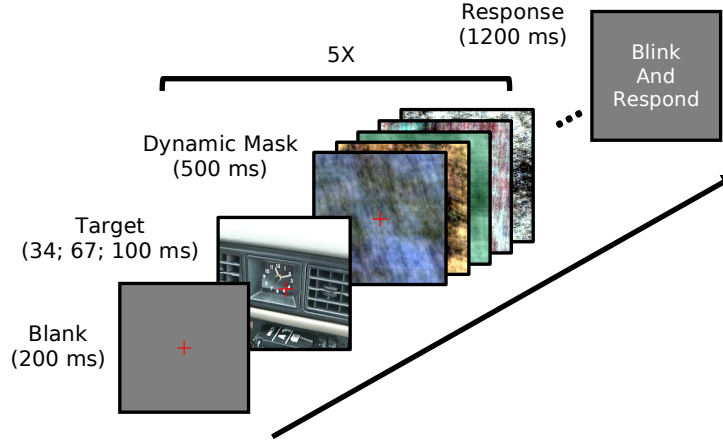


Figure 3. Schematic Overview of the EEG Experiment.

40% of the Challenge and Control images that exhibited the most substantial differences at that PT. During this selection process, we made sure to keep the same pairs of Control and Challenge images originally identified in the pairing procedure used to create the two groups of images (Boyanova, 2023). By including this step, we controlled for differences in human performance between images, thereby avoiding a potential confounding factor for the identification of Challenge and Control images. We included a total of 84 images depicting five objects, equally divided between the two groups. The distribution among objects was as follows: 12 for bird, 24 for book, 26 for bottle, 12 for clock, and 10 for elephant.

2.3.3 Experimental Design and Procedure

The experiment consisted in the presentation of a sequence of images along with a distractor detection task (**Fig. 3**). Participants viewed sequences of five images overlaid with a red fixation cross on a gray screen. The images were randomly presented for a duration of either 34, 67, or 100 ms. In 4% of the trials, instead of an image from the stimulus set, a catch trial depicting a paperclip was displayed for 200 ms within the sequence. After each image or catch trial, a dynamic mask appeared for a total duration of 500 ms. The dynamic mask was uniquely generated for each trial using the same procedure as in the behavioral experiment (**Fig. 1C**). A blank screen was presented for 200 ms between the end of the mask and the onset of the next image. At the end of each sequence, participants were instructed to first blink and then report with a keypress if a paperclip had appeared in the sequence. We used this detection task to ensure that participants were attentive throughout the experiment (hit rate = 94%, $SD = \pm 2.6$).

Each experimental image was displayed 30 times at each PT, resulting in a total of 7,560 trials per recording session. Each participant took part in a single experimental session lasting approximately 2 hours and 30 minutes. The session was divided into 33 blocks of 48

sequences, with each block lasting approximately 4 minutes and 30 seconds. The experiment was programmed in MATLAB (2019b) using Psychtoolbox (Brainard, 1997).

2.3.4 Data Acquisition and Preprocessing

We recorded EEG data with a 64-channel Easycap and a Brainvision actiCHamp amplifier (actiCHamp Plus, Brain Products GmbH, Gilching, Germany). Electrodes were arranged according to the standard 10–10 system. The data was recorded at a sampling rate of 1000 Hz, and online filtering was performed between 0.3 Hz and 70 Hz. We carried out offline data preprocessing in Python using the MNE package (Gramfort et al., 2013). First, we applied a low-pass filter of 40 Hz, and then we segmented the raw data into epochs of 550 ms, using a prestimulus window of 50 ms and a poststimulus window of 500 ms. Then, we performed baseline correction, by subtracting the mean voltage of the prestimulus window from the poststimulus window. The preprocessing of the data resulted in 30 trials for each image and PT per participant, which we used for further analysis.

2.3.5 Multivariate Pattern Analysis (MVPA)

At subject level, we conducted object decoding on the preprocessed EEG data using linear Support Vector Machine classifiers (SVM), implemented with the Python library Scikit-learn (Pedregosa et al., 2011). This analysis was performed separately for Control and Challenge images and for each PT.

Firstly, we downsampled the trials to 200 Hz to reduce computation time. Next, separately for each time point, we created 5 pseudo-trials by randomly grouping and averaging trials belonging to the same object. Due to the unequal number of images per object, we selected only 5 images, corresponding to the smallest number of images per object. Once obtained the pseudo-trials, we applied multivariate noise normalization (Guggenmos et al., 2018) to account for variations in electrode noise levels. We then trained and tested a one-versus-one SVM using the pseudo-trials, adopting a leave-one-out cross-validation approach. Finally, we repeated this entire process 100 times with a new random assignment of trials into pseudo-trials, and we averaged the results across these 100 repeats.

The classification results (percentage of decoding accuracy, 50% chance) were stored in a decoding matrix with dimensions 5 x 5 (object by object). Each cell in this matrix indicates the accuracy of the classifier in distinguishing between two objects. Subsequently, we calculated the average decoding accuracy across objects by averaging the lower triangle of the decoding matrix (excluding the diagonal) at each time point. This results in six time-

courses of decoding accuracy per subject – one for each experimental condition derived from the two groups of images and the three PTs.

Next, we conducted two types of analyses. In the first, we first averaged the decoding accuracy across participants and we then inspected the trajectories and peak latencies of the resulting time-course. In the second analysis, instead of computing the peaks on the decoding accuracy across subjects, we extracted peak latencies from each participant. Subsequently, we compared the means of these latencies across PTs, separately for Control and Challenge images.

2.4 Statistical Analysis

We estimated the means and confidence intervals of the behavioral metrics using bootstrapping (sampling with replacement) with 10,000 iterations.

In both the behavioral and RSA analyses, we utilized nonparametric permutation tests (Nichols & Holmes, 2002) to evaluate statistical significance. To test the difference in d' and RTs between Control and Challenge images, we randomly grouped Challenge and Control images into two sets, calculating the difference in their means for each metric. We iterated this process 10,000 times, resulting in a null distribution that we used to compute p-values for the empirical differences. This procedure was repeated independently for each PT.

In the RSA analysis, we employed a similar permutation approach for two different statistical assessments. In the first case, we tested whether a similarity measurement significantly differed from zero. In the second case, we examined whether the difference between two similarities significantly deviated from zero.

For the first test, the permutation procedure involved an initial step in which we shuffled the image labels of the behavioral RDM and then correlated this randomized RDM with the model RDM. We iterated this process 10,000 times to generate a null distribution and calculated p-values for the empirical correlations. In the second case, we utilized the same approach as described earlier, with the exception that we applied randomization of behavioral RDM and correlation with model RDM to both similarities we intended to compare. Next, we calculated the differences between the correlations we obtained, and, as before, we iterated the process 10,000 times, calculating p-values for the empirical differences between similarities.

A significance threshold of 5% was consistently applied in all tests, utilizing a two-tailed test approach. To control for familywise error, we employed Bonferroni correction.

No statistical analysis was conducted on the decoding results from the EEG experiment, as it served as a pilot study to gather preliminary data.

3. Results

3.1 Behavioral Results

We used d' and RTs to assess the behavioral performance of participants in recognizing a depicted object among two alternatives (**Fig. 1B**).

Due to the stronger reliance on recurrent processing that we assumed for Challenge images, we anticipated that those images would exhibit a more pronounced impairment of d' at shorter PTs compared to Control images. We expected this based on the assumption that the shorter duration between the image's onset and the masking's onset would not allow a full unfolding of the recurrent processing. In contrast, we expected that there would be enough time for this to happen at longer PTs. We also anticipated longer RT for Challenge images, as the engagement of recurrent processing should increase their processing time.

Furthermore, we compared our results at 100 ms PT with the behavioral outcomes from Boyanova (2023), where a study involving a similar target discrimination task as ours was conducted, but without the implementation of backward masking.

3.1.1 Recognition of Challenge Images is Impaired at 67ms Presentation Time

The d' results (**Fig. 4A**, or Appendix A1 for details) revealed a linear decrease in performance at shorter PTs, specifically between 200 ms and 67 ms, with an even steeper decrease observed between 67 ms and 34 ms.

When comparing d' of the two image groups, we found a difference ($0.1 d'$, $p = .04$) at 67 ms with reduced d' for Challenge images and no significant difference at longer PTs (all $p > .05$, for details see inserts in **Fig. 4A**). Additionally, no significant difference was observed at a PT of 34 ms ($p = .12$). Those results indicates that, at 67 ms PT, participants had more difficulty in recognizing Challenge images compared to Control images, while images from both groups were recognized equally well at longer PTs. Considering that we expected recurrent processing to be mostly disrupted at shorter PTs, this result aligns with our expectation suggesting a stronger reliance of Challenge images on recurrent processing.

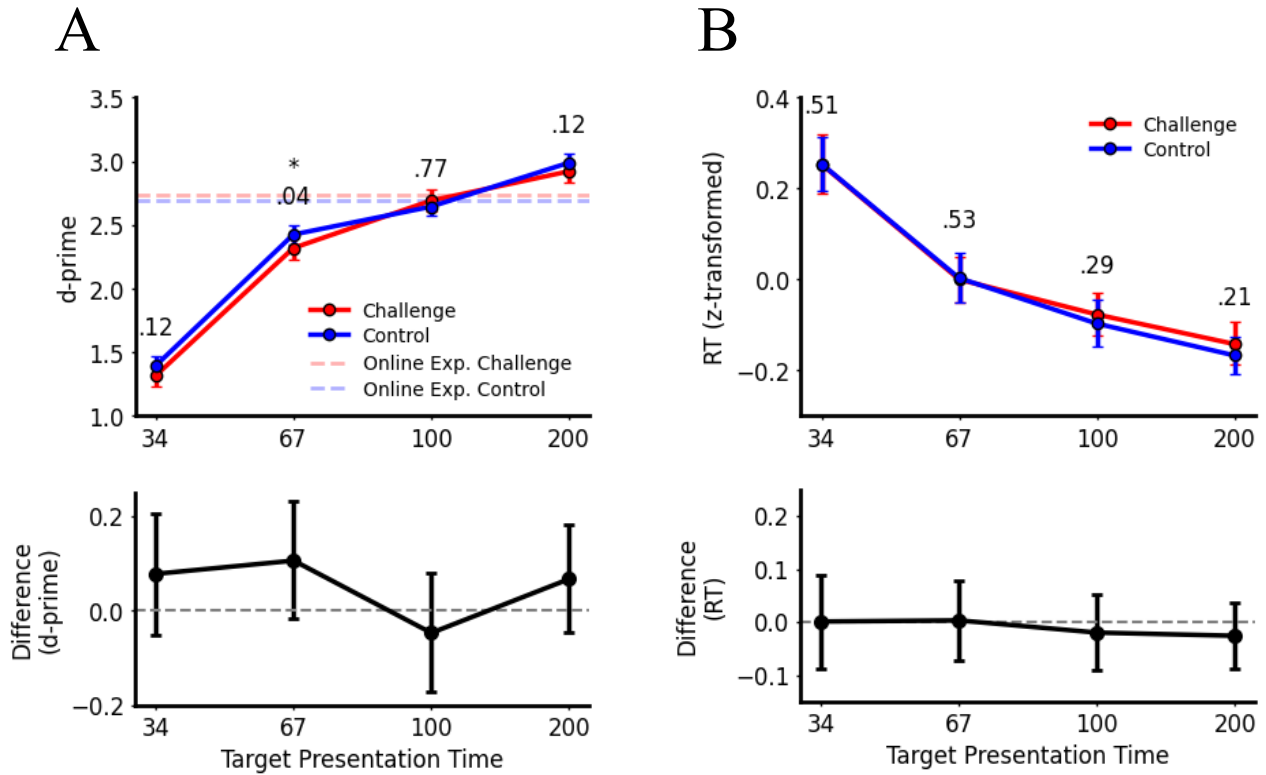


Figure 4. Behavioral Results. **A)** Results of the d' analysis. The blue line represents Control images, while the red line represents Challenge images. Dashed lines indicate the results of the online behavioral experiment, used for categorizing images into Control and Challenge. The online experiment implemented the same object recognition task with images presented for ~100 ms and no backward mask. Error bars in all plots represent 95% bootstrapped confidence intervals, and annotations indicate p-values from two-tailed permutation tests. **Bottom)** Differences between Control and Challenge images depicted on the upper plot are reported in the lower plot. **B)** Results of the RTs analysis. The blue line represents Control images, while the red line represents Challenge images. **Bottom)** Same as in A.

Furthermore, at a 100 ms PT our results exhibit performance comparable to that observed in Boyanova (2023) (**Fig. 4A, dashed lines**), who conducted an experiment with the same design and stimuli as ours, albeit using only a 100 ms PT and without backward masking. This finding provides converging evidence that backward masking does not influence the recognition of Challenge images at 100 ms PT, as both groups of images display the same level of d' , whether with or without backward masking

3.1.2 No Significant Differences in Reaction Times across Presentation Times

We compared the RTs between Control and Challenge images at each PT (**Fig. 4B** or Appendix A2 for details). It's important to note that our focus was on comparing RTs within

the same PT for different image groups, rather than across different PTs. Our experimental design was not suited for the latter analysis because varying PTs could introduce biases into RT durations. For instance, when an image is presented for 200 ms, shorter RTs might be expected due to more available time for providing a behavioral response compared to a PT of 34 ms. However, by comparing RTs of Control and Challenge images at the same PT, we avoided this bias, as both groups were presented for the same duration.

We observed longer RTs for Challenge images at PTs of 100 ms and 200 ms compared to Control images, although these differences were not statistically significant (all $p > .05$, for details see inserts in **Fig. 4B**). Despite the lack of statistical significance, the observed pattern aligns with our expectations of Challenge images relying more on recurrent processing. Indeed, we expected that the engagement of recurrent processing would result in increased RTs, as this processing has been shown to occur later in time compared to feedforward processing (Lamme & Roelfsema, 2000; Wyatte et al., 2014). Most important, we observed longer RTs for Challenge images only at longer PTs, aligning with our expectation that recurrent processing would be less disrupted by backward masking at longer PTs.

3.2 Computational Modeling

We employed RSA to compare the internal representations of various CNN architectures to human behavior.

Our main objective was to investigate whether models incorporating recurrence-like features, such as recurrent connections or increased depth, exhibited enhanced similarity to human behavior. To assess this, we first examined the evolution of similarities of Control and Challenge images across models' layers, testing for significant deviations from zero. AlexNet served as our benchmark, as it was used for creating the image groups (Boyanova, 2023). For this model, we anticipated that, as the depth of layers increased, there would be no increase in similarity for Challenge images, as these images were categorized as such due to AlexNet's inability to recognize them. Conversely, we expected an increase in similarity for both groups of images in models implementing recurrent-like features. This expectation was based on the hypothesis that adding recurrence-like features may enable the models to partially capture human recurrent processing (Spoerer et al., 2017).

Furthermore, we identified the model with the highest similarities for both Control and Challenge images, aiming to determine the most effective CNN architecture in capturing

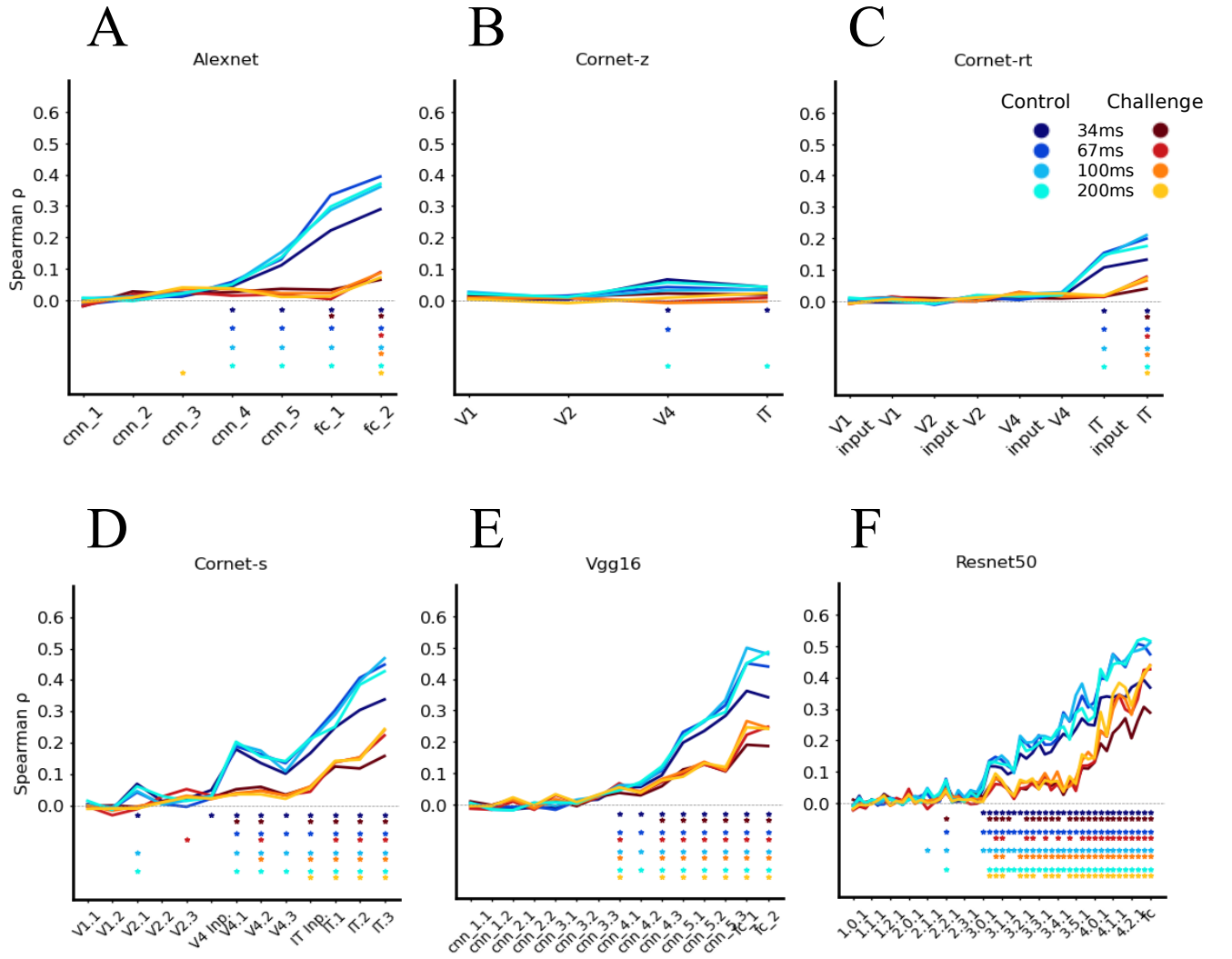


Figure 5. Fit to Human Behavior at Different PTs of Various CNN Architectures. **A)** Spearman rank correlations between AlexNet and human behaviors at different PTs. Shades of blue represent Control images, while shades of red represent Challenge images. Different shades indicate various PTs, with darker shades referring to shorter PTs and lighter shades to longer PTs. Asterisks at the bottom indicate correlations significantly different from zero (two-tailed permutation tests, $p < .05$). **B)** Same as in A but for Cornet-z. **C)** Same as in A but for Cornet-rt. **D)** Same as in A but for Cornet-s. **E)** Same as in A but for VGG16. **F)** Same as in A but for ResNet50.

human object recognition across all images. To achieve this, we extracted similarities from the last layer of each model for both groups of images. Subsequently, we tested whether differences between these similarities significantly deviated from zero, aiming to examine whether a model achieved comparable levels of similarities for both groups.

3.2.1 Models Implementing Recurrence-Like Features Show Higher Similarities to Humans for Both Control and Challenge Images

We first examined the trajectories of similarities across layers of all implemented models, evaluating whether they significantly deviated from zero (**Fig. 5A-F**).

Overall, we found a consistent trend of increasing similarities for Control images as the layers got deeper. This was observed for all models, with the exception of CorNet-z. In contrast, an increase in similarities for Challenge images was observed exclusively in CorNet-s, VGG16, and ResNet50, which are models implementing recurrence-like features.

These results align with our expectations. Firstly, we found that AlexNet exhibited an increase in similarities for Control images only. This confirmed that the internal representation of the two groups of images identified using Kar’s approach displayed a different alignment to human behavior.

Secondly, our results highlight that only models incorporating recurrence-like features demonstrate increasing similarities for Challenge images. This suggests these models might capture some aspects of human recurrent processing.

Interestingly, we also observed a persistent gap between the two image categories across all models, consistently resulting in lower similarities for Challenge compared to Control images.

3.2.3 ResNet50 Demonstrates the Strongest Alignment with Human Behaviors

To further characterize our results, we sought to identify the model with the best alignment with human behavior. We first extracted similarities of Control and Challenge images from the final layer of each model, at each PT (**Fig. 6, top**). Then, we computed their difference (Control minus Challenge, **Fig. 6, bottom**) and conducted significance tests. We carried out this statistical assessment to determine whether each model displayed consistent similarities between the two groups of images. Through this test, our aim was to further evaluate our models, as our belief is that a model fully capable of capturing human object recognition should ideally demonstrate equally high similarities for both groups of images.

Among the models we implemented, we expected CORNet-S (Kubilius et al., 2018) to outperform the rest. This anticipation was based in earlier research showing that CORNet-S achieved comparable performance in behavioral predictivity to a very deep CNN like ResNet101 (He et al., 2016), when evaluated on Brain-Score (Schrimpf et al., 2018), a benchmark measuring a neural network's ability to predict neural activity and human behavioral performance.

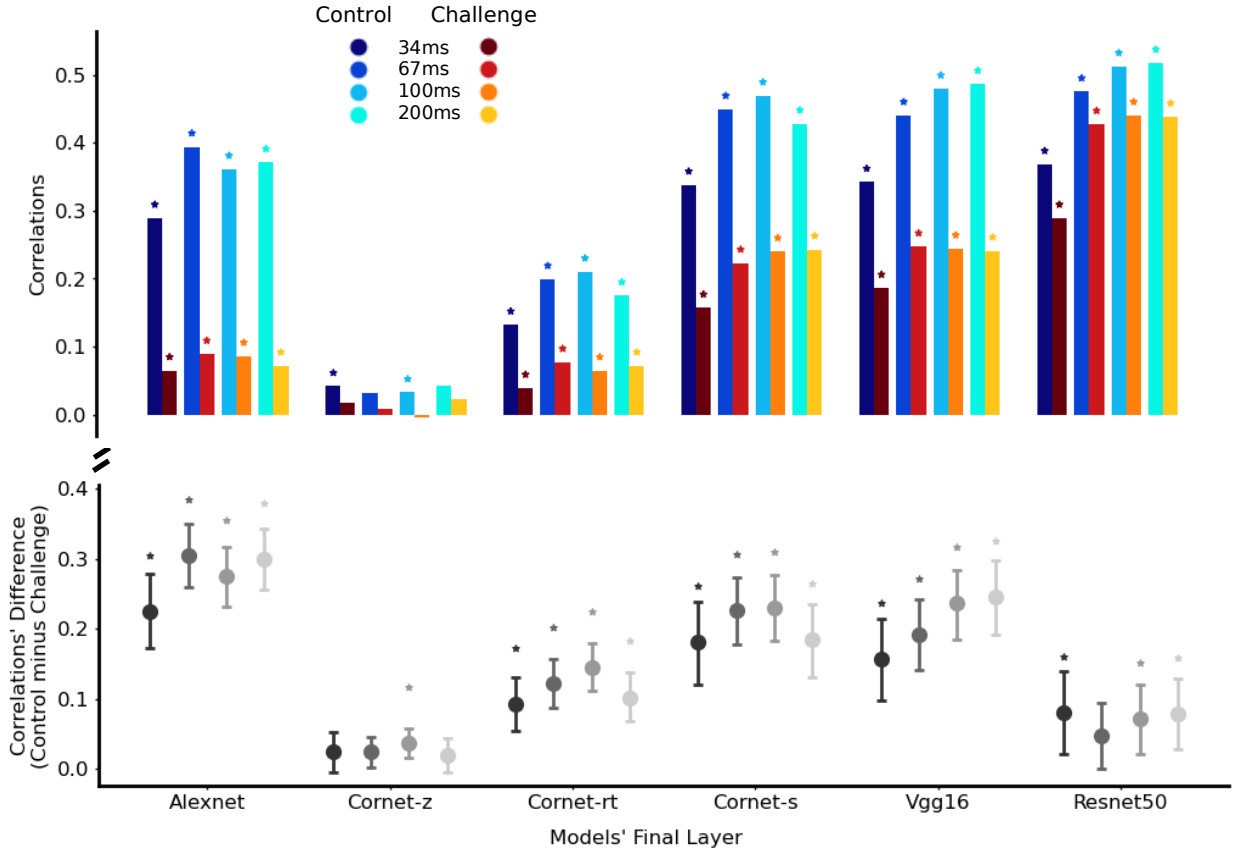


Figure 6. Difference between Control and Challenge Images. Top) Spearman rank correlations between models' final layer and human behaviors at different PTs. Shades of blue represent Control images, while shades of red represent Challenge images. Different shades indicate various PTs, with darker shades referring to shorter PTs and lighter shades to longer PTs. Asterisks at the top indicate correlations significantly different from zero (two-tailed permutation tests, $p < .05$). **Bottom)** Correlations' difference between Control and Challenge images (Control minus Challenge) corresponding to the correlations shown in the upper plot. Asterisks at the top indicate significantly different correlations (two-tailed permutation tests, $p < .05$).

Others than expected, our results indicate that ResNet50 displayed the highest similarities for both groups of images, accompanied by the smallest difference between them. Notably, the higher similarities for the Challenge images are the most noticeable differences from the other models, suggesting that ResNet50 is particularly effective in capturing human behaviors for those images. Although smaller than in other models, the similarities' differences remained significant across all PTs (all $p < .05$) except for 67 ms PT ($p > .05$, **Fig. 6, bottom**).

However, this last result is likely attributed to a decrease in the similarity of Control images rather than an increase in Challenge image similarity.

In summary, our results identify ResNet50 as the most effective model in capturing human behaviors, showing especially enhanced performance with Challenge images compared to other models. Furthermore, we observed a persistent gap in similarities between Control and Challenge images, with Control images displaying higher similarities across all models.

3.3 Neural Decoding

We conducted a pilot experiment implementing EEG and MVPA. In the context of MVPA analysis, a high object decoding accuracy signifies a high discriminability of the object identity from the neural activity (Cichy et al., 2014) and peaks in decoding accuracy indicate the timepoints at which object identity discriminability is highest. Our primary goal was to investigate how variations in PTs, along with the presence of backward masking, influence the decoding accuracy in both Control and Challenge images.

We conducted two distinct analyses. In the first analysis, we averaged the decoding accuracies across subjects and examined how the resulting decoding curves, as well as the decoding peak latencies, differed between Control and Challenge images and different PTs. In the second analysis, we extracted peak latencies of Control and Challenge images from the decoding accuracies of each subject, and we explored their average variations at different PTs.

3.3.1 Object Decoding Revealed Higher Peaks for Control Images and Lower but More Constant Trajectories for Challenge Images

Previous studies investigating the latencies of feedforward and recurrent processing have shown that recurrent processing exhibits longer latencies compared to feedforward processing (Lamme & Roelfsema, 2000; Wyatte et al., 2014). Building on these previous findings, we expected that the decoding accuracy of Challenge images would peak at later timepoints compared to Control images. This would indicate the need of additional time for the processing of Challenge images, suggesting a stronger involvement of recurrent processing.

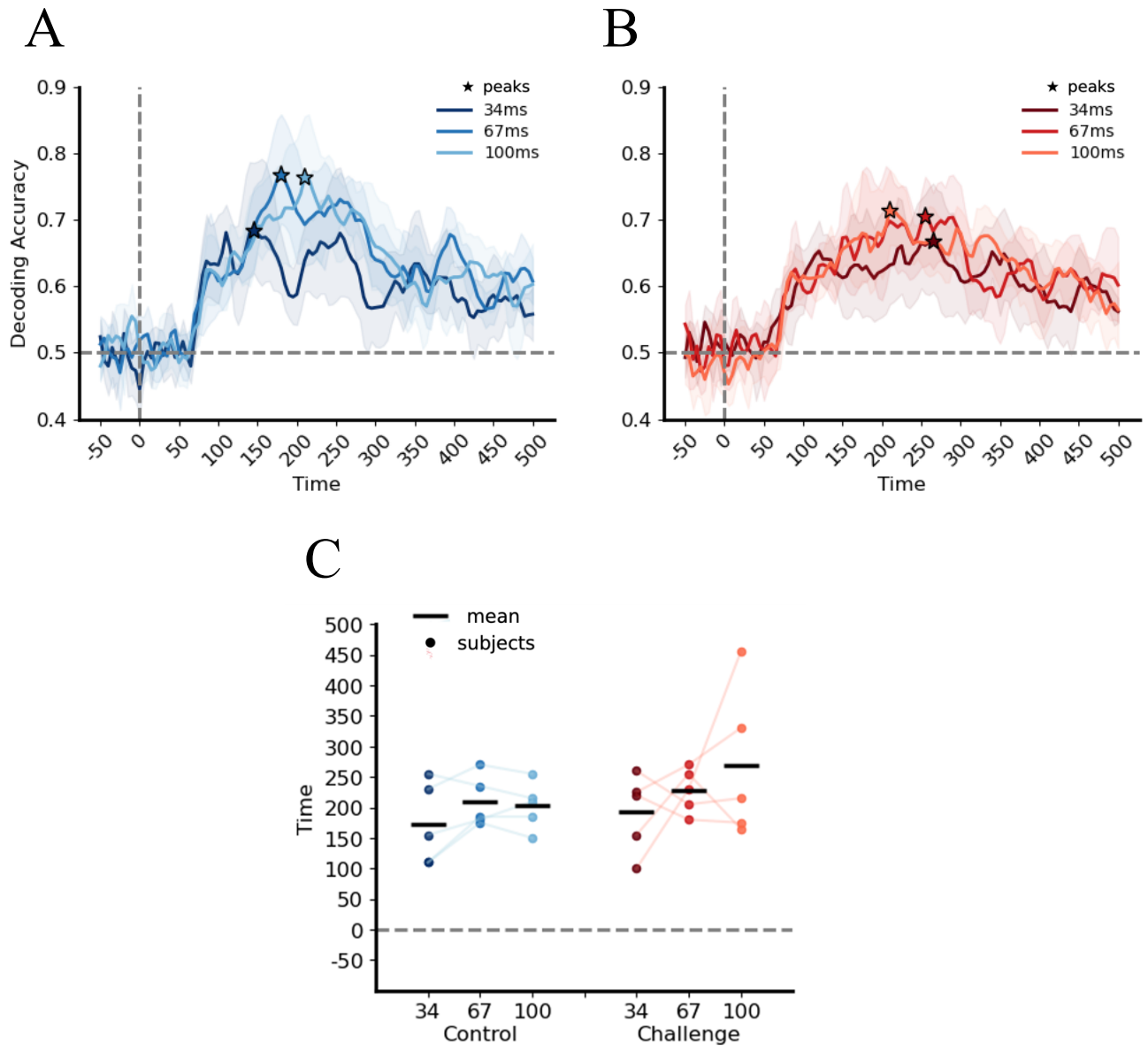


Figure 7. Object Decoding for Control and Challenge Images. **A and B)** Time course of object decoding for Control and Challenge images respectively. Different shades indicate various PTs, with darker shades referring to shorter PTs and lighter shades to longer PTs. Asterisks at the bottom indicate peaks in decoding accuracy. The vertical gray line at 0 ms marks the images onset, while the horizontal line indicates chance level. **C)** Peaks latencies of object decoding by subjects. Dots represent peaks for single subjects, and lines connect dots belonging to the same subject. Black lines indicate averaged latencies. Shades of blue represent Control images, while shades of red represent Challenge images. Different shades indicate various PTs, with darker shades referring to shorter PTs and lighter shades to longer PTs. The vertical gray line at 0 ms marks the images onset.

The results of the decoding accuracy across subjects, align with our expectations. For Control images, we observed that at PTs of 67 and 100 ms, the object decoding trajectories peaked between ~180–220 ms (**Fig. 7A**) and rapidly decline at later timepoints. In contrast, trajectories for Challenge images peaked later, between 250–270 ms (**Fig. 7B**), and exhibit a more constant course across the time window.

The earlier peaks displayed by Control images align with the engagement of faster feedforward processing, and the steep decline observed at later timepoints further supports this, indicating that later processing is less involved in the processing of object identities. In contrast, the later peaks of Challenge images fit better with a stronger engagement of slower recurrent processing.

We also noticed that, at the 34 ms PT condition, both groups of images exhibited a significant reduction in their decoding trajectories (**Fig. 7A and 7B**). This observation aligns with the decline in behavioral performance observed in the behavioral experiment at this PT (**Fig. 4A**), indicating that such a short PT impair the processing of both groups of images.

3.3.2 Averaged Peaks by Subjects: Earlier Latencies at Shorter PTs for Challenge Images

In addition to examining the decoding accuracy across subjects, we also investigated the mean latencies of the decoding accuracy peaks computed for each subject (**Fig. 7C** or Appendix A3 for details). With this analysis, we aimed to further assess the impact of recurrent processing disruption on the two groups of images. We anticipated that a greater dependence on recurrent processing would result in the presence of later peaks, and disruptions of this processing would cause a shift of those peaks to earlier time points. Conversely, we expected that images relying more on feedforward processing would exhibit earlier peaks and would not be strongly affected by impairments of recurrent processing.

Observation of the pattern in the mean peak latencies support our hypothesis. Firstly, consistent with our findings on the decoding accuracy across subjects (**Fig. 7A-B**), we observed later latencies for Challenge images (mean = 268 ms) compared to Control images (mean = 203 ms) at 100 ms PT. Secondly, the mean peak latencies of Challenge images exhibited a shift to earlier time points at shorter PTs (**Fig. 7C red**), while no shift was observed for Control images (**Fig. 7C blue**). Both of these results align well with the hypothesis of a stronger engagement of recurrent processing for the recognition of Challenge images.

4. Discussion

In this study, we extended previous findings on the role of recurrent processing in object recognition (Oyarzo et al., 2022; Rajaei et al., 2019). We showed that the disruption of recurrent processing more strongly impairs the processing of Challenge images compared to Control images. To achieve this, we selectively disrupted recurrent processing through backward masking and variations in PTs (Fahrenfort et al., 2007; Kar et al., 2019), evaluating how this disruption affected the recognition and decoding accuracy of Control and Challenge images. We also explored the similarities between human behaviors and the internal representations of various CNN models.

We highlight three key findings. First, we observed a more pronounced impairment in the recognition of Challenge images under conditions where recurrent processing was expected to be more disrupted. Secondly, we observed that models implementing recurrence-like features demonstrated better alignment with human behaviors, with the most noticeable difference from other models being an increase in similarities for Challenge images. Finally, in our neural decoding analysis, we observed constant object discriminability in Challenge images at later timepoints, which diminished with greater disruption of recurrent processing.

4.1 Evidence of Impaired Recognition of Challenge Images after Disruption of Recurrent Processing

We reasoned that if recurrent processing supports the recognition of Challenge images, its disruption should lead to impaired recognition for those images. As expected, at shorter PTs, where recurrent processing disruption was stronger, Challenge images showed reduced recognition compared to Control images, with no difference at longer PTs. These results suggest a critical role of recurrent processing in Challenge images recognition, aligning with previous studies reporting impaired recognition of occluded (Rajaei et al., 2019; Tang et al., 2014) or degraded images (Sejdel et al., 2021; Tang et al., 2018; Wyatte et al., 2012) following backward masking.

Contrary to expectations, there was a similar decrease in object recognition for both image groups at the shortest PT, potentially indicating a floor effect. To make sense of this, we speculated that excessively short PTs might have impaired feedforward processing. Interestingly, unlike our findings, previous studies using backward masking achieved higher recognition performance at similar or even shorter PTs (Sejdel et al., 2021; Tang et al.,

2018). However, it's worth noting that those studies used grayscale synthetic images while we employed colored natural images, posing a challenge for result comparison due to potential differences in the neural processing of the stimuli (Bramão et al., 2011).

4.2 CNNs Incorporating Recurrence-Like Features Display Enhanced Similarities with Human Behaviors

Expanding the results of Kar et al. (2019), which demonstrated the capability of shallow feedforward CNNs to capture the neural processing involved in the recognition of Control images in monkeys, we hypothesized that models implementing recurrence-like features (such as deeper models or those with recurrent connections) would also be capable to capture recurrent processing that we assume are needed for recognizing Challenge images in humans. We tested this hypothesis by comparing models' internal representations with human behavior using RSA. Based on our expectations, we anticipated that shallow feedforward models would demonstrate higher similarities with human behavior exclusively for Control images, while models with recurrent-like features would exhibit higher similarities for both image groups.

Most of our results aligned with our expectations. We observed that shallow feedforward CNNs showed increased similarities, as the layers got deeper, only for the Control images. In contrast, consistent with previous findings (Nayebi et al., 2018), we observed that models with recurrence-like features became more aligned with human behavior for both groups of images. These findings potentially indicate that incorporating recurrence-like features enables CNN models to partially captures human recurrent processing.

However, our results also revealed that, even in models with recurrence-like features, Challenge images exhibited lower similarities compared to Control images. This difference could stem from various factors. One possibility is that the chosen CNN architecture might not be optimal for fully capturing human recurrence. Alternatively, the gap may be attributed to intrinsic differences between humans and CNNs in image processing. Several studies, indeed, have pointed out potential limitations in using CNNs to model human object recognition (Wichmann and Geirhos, 2023 for a review), such as the reliance on texture vs. shape (Baker et al., 2018; Singer et al., 2022), susceptibility to adversarial attacks (Szegedy et al., 2014) and susceptibility to changes in viewpoints (Dong et al., 2022).

4.3 Disrupting Recurrent Processing Affects Decoding Accuracy of Challenge Images

Using EEG and MVPA, we presented additional evidence indicating that Challenge images rely more on recurrent processing than Control images.

First, our results showed constant object discriminability at later timepoints for Challenge images, while Control images displayed early peaks followed by a sharp decline. As recurrent processing typically occurs later compared to feedforward processing (Lamme & Roelfsema, 2000; Wyatte et al., 2014), we argue that the presence of object information at later time points for Challenge images reflects an engagement of recurrent processing.

Second, we showed that disruption of recurrent processing affects more Challenge images, implying a stronger reliance on this processing. We derive this interpretation from the observation of a shift to earlier time points in the mean peak latencies of Challenge images, a pattern not observed for Control images.

Collectively, our results substantiate the hypothesis of a stronger involvement of recurrent processing for the recognition of Challenge images, aligning with studies exploring the role of recurrent processing with occluded stimuli (Rajaei et al., 2019) or partial stimuli (Tang et al., 2014, 2018). Extending previous research, we provide converging evidence that Kar's results can be generalized to humans (Oyarzo et al., 2022) and to natural images (Boyanova 2023).

However, identifying the origin of the recurrent processing supporting the recognition of more challenging images remains an ongoing endeavor. A recent study (Kar and DiCarlo, 2021) showed that inactivating the vLPFC in monkeys impairs recognition of Challenge images, making their processing in the IT cortex more akin to Control images. This may suggest that, in our study, backward masking caused a disruption of recurrent processing from the vLPFC. Studies that implement disruption of recurrent processing and spatiotemporal investigations are needed to explore this further.

4.4 Limitations and Future Directions

We highlight four limitations in our study. A first limitation concerns the small effects found in the behavioral experiments. One possibility is that our backward masking is affecting both recurrent and feedforward processing, as proposed in some models of visual masking (Macknik & Martinez-Conde, 2007), thereby reducing the recognition of both image groups. To address this issue, different masking approaches could be systematically examined in future studies.

Second, inconsistencies between our behavioral results and those of Kar (2019) pose a limitation. Apart from larger differences in d' , Kar observed significant disparities in d' up to 134 ms PTs. One potential explanation could be intrinsic differences in neural processing between monkeys and humans, although a study has demonstrated similarities in neural representations of object recognition between them (Rajalingham et al., 2015). Alternatively, the different images used might be a factor; Kar employed grayscale synthetic images, while we utilized colored natural images (Mehrer et al., 2021). Replicating the current study with Kar's images would help to address this.

A third limitation relates to the comparison of different CNN architectures in our RSA analysis. Our goal was to offer a broad examination of various benchmark models, leading us to compare models differing along multiple dimensions. However, this approach hinders the identification of which specific feature contribute to better capturing human recurrent processing. To address this limitation, future studies could implement parametrically matched models, enabling systematic comparison of models differing along well-specified dimensions.

Lastly, a small sample size in the EEG experiment limits the strength of conclusions drawn from these results. A full-scale study would be needed to make stronger claims about the effect of recurrent processing disruption on the decoding accuracy trajectories and peak latencies.

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Appendix: Supplementary Tables

Table. A1 Statistical Details for the d' Results

<i>Condition</i>	<i>Control</i>	<i>Challenge</i>	<i>difference (Con. – Chall.)</i>	<i>p-value</i>	<i>95% CI</i>
34 ms	1.39	1.31	.08	.12	[-.05 .20]
67 ms	2.42	2.31	.11	.04*	[-.01 .23]
100 ms	2.64	2.69	-.05	.77	[-.17 .08]
200 ms	2.98	2.92	.06	.12	[-.04 .18]

Table. A2 Statistical Details for the RTs Results (z-transformed)

<i>Condition</i>	<i>Control</i>	<i>Challenge</i>	<i>difference (Con. – Chall.)</i>	<i>p-value</i>	<i>95% CI</i>
34 ms	.25	.25	0	.51	[-.08 .08]
67 ms	0	0	0	.53	[-.07 .07]
100 ms	-.09	-.07	-.02	.29	[-.09 .05]
200 ms	-.16	-.14	-.02	.21	[-.09 .03]

Table. A3 Decoding Peaks Latencies by Subjects (Average)

<i>Latencies (ms)</i>		
<i>Conditions</i>	<i>Control</i>	<i>Challenge</i>
34ms	172	192
67ms	209	228
100ms	203	268